# Précis and autocatpath: a citation-grounded, untiring agentic loop for catalyst discovery

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Précis [1] is an open-source agentic research system built as an *untiring collaborator*: it never sleeps, works from real feedback, and lets no claim stand without a precise citation — a specific paragraph in a real publication that supports it, so hallucination is designed out rather than apologized for. Papers arrive automatically from open-access sources and continuous citation-graph exploration; the corpus currently holds over 22,000 fully ingested papers (33,000 tracked), broken into roughly 3 million searchable chunks. Precomputed keywords, summaries, and a clustered dynamic table of contents let language-model agents locate very specific claims token-efficiently instead of reading whole papers. Agents carry real tools — symbolic math, calculators, search backends, and simulators — and tasks can park themselves until a paper is ingested or a human answers, then resume on their own.

The computational arm is autocatpath [2], a standalone GPLv3 reaction-pathway explorer for catalyst surfaces driven by machine-learned interatomic potentials. Given an environment, a substrate, and a target, it builds the reaction network (curated templates or rule-based autodetection), relaxes every intermediate, finds barriers with climbing-image NEB [3], and reports energies with honest uncertainty pooled across random seeds *and* across potentials (MACE [4], CHGNet [5], FAIRChem/UMA, GRACE). Computational-hydrogen-electrode limiting potentials [6] come for free from energies already computed, and a relax-only screening mode ranks candidates on thermodynamics before any barrier is bought.

The *quest loop* couples the two into a perpetual autonomous discovery process. A discovery agent owns all chemistry and holds one proposal in flight; a code-driven tier ladder promotes candidates from screening through NEB to a verify tier under capped budgets, and results rank on a Pareto frontier over the quest's objective axes (barrier, material cost, stability). Untrusted measures — unconverged NEB images, detached adsorbates, symmetry-twin pairs whose barriers disagree — can never rank or graduate, but stay visible as provisional bands so the loop is never blind to its own measurements. Conjectures are minted as typed hypothesis findings that accumulate literature evidence for and against between compute steps, and a refuted hypothesis converts into a negative ruling that is excluded from re-proposal by default. The governing design principle is to substitute thinking for compute — one hour at the library is worth a month in the simulator — so work runs as tight idea–probe–evaluate cycles rather than blind sweeps. We demonstrate the loop on NO → $NH_3$ over Pd-based catalysts (Figure 1).

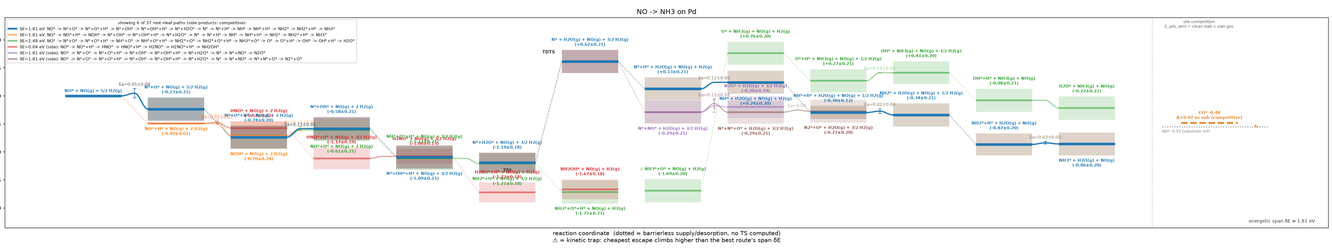

Figure 1: autocatpath output for NO → $NH_3$ on Pd (EMT demonstration backend): span-ranked competing routes at full gas-ledger composition, transition states with $E_a$ and uncertainty bands, kinetic-trap flags, and the CO site-competition panel — from a single `autocatpath run`.

## References

[1] R. Stamm, *Précis*, https://github.com/retospect/precis-mcp (2026).
[2] R. Stamm, *autocatpath*, https://github.com/retospect/catpath (2026).
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[4] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, *Adv. Neural Inf. Process. Syst.* **35**, 11423 (2022).
[5] B. Deng *et al.*, *Nat. Mach. Intell.* **5**, 1031 (2023).
[6] J. K. Nørskov *et al.*, *J. Phys. Chem. B* **108**, 17886 (2004).